Uncertainty-Aware Routing Needs an Ensemble You Do Not Have

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Abstract

Escalating a query to an expensive model pays off only when the cheap model's error is *reducible*, so uncertainty-aware routers decompose predictive uncertainty and route on the reducible part. Two 2026 results establish the pieces this rests on: the mutual-information estimate of epistemic uncertainty is not the reducible kind, and the irreducible term cancels from the escalation decision. Both assume the estimator one would like — independently seeded ensemble members, or multiple independent annotations per input. We ask what the estimator you can actually build does instead, in a synthetic video-forecasting world where the true conditional P^* is exact in closed form, so every estimate can be compared against the quantity it claims to estimate. (i) As members go from independent to identical at fixed per-member error, the ensemble's true error rises $10.0\times$ while its estimate falls to zero; they cross near $\rho = 0.5$, and the estimator's advantage over random routing is gone by $\rho = 0.75$. (ii) Whether confidence routing decays or inverts below random is decided by *why* the cheap model is worse: benign under missing evidence, catastrophic under miscalibration, floor-invariant where the estimate is well-posed. (iii) Video has a second lever, and 92%–98% of the gain from extending the observed prefix is exactly the term that cancels from model escalation. Everything reproduces over 5 seeds and two structurally different worlds. No vision-language model was run, and nothing here is evidence about real video.

1 Introduction

A forecaster sees a prefix of video and reports a distribution over what happens next. For a forecast Q on a scene whose true conditional is P^*, the expected log loss splits exactly:

$$\mathbb{E}_{y\sim P^*}[-\log Q(y)] \;=\; \underbrace{H(P^*)}_{\text{irreducible}} \;+\; \underbrace{\mathrm{KL}(P^* \,\|\, Q)}_{\text{the forecaster's}} \tag{1}$$

This is an identity. $H(P^*)$ is a property of the world and the prefix: no forecaster reduces it and every forecaster pays it. $\mathrm{KL}(P^* \,\|\, Q)$ is a property of the forecaster, zero exactly at Bayes-optimality, and it is the entire quantity a better model could recover. Escalating one scene from a cheap forecaster Q_c to an expensive Q_e therefore changes expected loss by

$$\big[H(P^*) + \mathrm{KL}(P^* \,\|\, Q_c)\big] - \big[H(P^*) + \mathrm{KL}(P^* \,\|\, Q_e)\big] \;=\; \mathrm{KL}(P^* \,\|\, Q_c) - \mathrm{KL}(P^* \,\|\, Q_e) \tag{2}$$

The irreducible term cancels exactly. But the total predictive entropy $H(\bar{Q})$ that confidence-based routing ranks by *contains* that term, and is often dominated by it.

What is already established, and is not claimed here. Young [9] proves that the standard definition of epistemic uncertainty and its standard mutual-information measure [3, 5] are extensionally inconsistent, and that ensemble disagreement "collapses to zero beneath a positive truth under consistent training". Peale et al. [7] state the cancellation in (2) in their introduction, build a router that decomposes total uncertainty via higher-order predictors [2], and show total-uncertainty routing underperforming a pointwise-optimal policy. So the inequality between disagreement and forecaster error, the cancellation, and the failure of total-uncertainty routing are *not* contributions of this paper. We take them as the setting.

What neither covers. Both assume the estimator one would like. Young [9] assumes independently seeded members; Peale et al. [7] require multiple independent annotations per input plus a higher-order calibration guarantee. The practical substitute is an ensemble of similarly pre-trained models, or repeated calls to one model, whose errors are correlated by construction. That correlation is the

free parameter neither paper sweeps, and it is the one a deployment does not get to choose. Relatedly, Kirsch [6] reports epistemic estimates collapsing as model *complexity* rises, by implicit ensembling — a different cause, and without a routing experiment or a known ground-truth epistemic error to compare against.

Contributions. Three, each measured against a closed-form P^*, against an explicit random-routing baseline, in two structurally different worlds: (i) a ρ*-parameterised* link from ensemble-member correlation to routing performance, in which the estimate and the error it stands in for cross (§3); (ii) a *mechanism map*, in which the cause of the cheap model's deficit — not the irreducible floor — decides whether confidence routing decays or inverts (§4); and (iii) the *prefix lever*, which is specific to video and whose gain is almost entirely the term that cancels from model escalation (§5). Everything is re-run across seeds and bootstrapped (§6). The world is synthetic, no vision-language model was called and no video corpus was used; §7 says what that costs.

2 An instrument in which the floor is known

P^* is not observable on real video: nobody knows the true distribution of the future given a clip, so $H(P^*)$ there can only be assumed. Every empirical claim about aleatoric uncertainty on real video is therefore a claim about an estimator whose target is unavailable for comparison. We work where the target exists.

A scene carries latent cues z with finite alphabets and independent priors. The outcome is drawn from $P(y \mid z)$, with logits linear in the cues (world JUNCTION) or carrying pairwise interactions (world KITCHEN), softened by a temperature τ. A prefix reveals a subset of the cues, recorded by an observability mask m drawn independently of z, so the best any forecaster can do is $P^*(y \mid z_m) = \sum_{z_h} P(z_h)\, P(y \mid z_m, z_h)$ — an exact finite sum over 81 joint cue states and 16 masks. τ is the only knob on the irreducible floor, which runs 0.72 to 1.33 nats against a ceiling of $\log 4 = 1.3863$; prefix length is a second, used in §5.

Forecasters are chosen so their behaviour is known in closed form before they run: Bayes-optimal (epistemic exactly zero), uniform (expected loss exactly $\log K$), marginal (excess loss exactly $I(y; z_m)$), partial-evidence, tempered, logit-noise with an ensemble-correlation parameter ρ, and cue-dropout. That is what makes the instrument checkable rather than merely plausible: 7 validation checks, each with an answer fixed by the mathematics rather than by a previous run, pass in *both* worlds — Bayes-optimal epistemic error 0 exactly, the marginal forecaster's excess loss 0.382810 against an enumerated 0.383812, and an evidence ladder matching every rung of its closed form (Appendix A).

Metric. We report the Spearman correlation between a policy's score and the true escalation gain of (2), per scene: it is zero when a policy carries no signal, and a *negative* value has a deployment meaning — the policy is worse than routing at random, spending the budget on the scenes with the least recoverable error. Budgeted curves normalised so random scores 0 and the oracle 1 are in Appendix D and agree on every sign below. Every number comes from 2018 held-out scenes and 12-member ensembles.

3 Correlated members: the estimate and the error move in opposite directions

The probe holds each member's own marginal noise scale fixed while sweeping the fraction ρ of the logit perturbation that members share. A single member's error therefore does not change with ρ; only the ensemble's ability to see it does. That invariance is what makes the sweep readable, and it is asserted by a test.

Table 1 reads in three parts. A single member's error is flat ($0.1569 \to 0.1491$). The deployed ensemble's *true* error rises $10.0\times$ ($0.0148 \to 0.1491$), because averaging cancels only the error the members do not share. The *estimate* falls to 3.7×10^{-19}, because it measures disagreement and identical members do not disagree. At $\rho = 0$ the estimate overstates the ensemble's error by $8.1\times$; at $\rho = 1$ it reports zero for an ensemble carrying 0.1491 nats of recoverable error. The two quantities cross near $\rho = 0.5$, so there is no correlation level at which the estimate is merely conservative.

Table 1: As ensemble members go from independent to identical at fixed per-member error (JUNCTION, τ=0.9). Column 1 is the control. The deployed ensemble's true error rises 10.0×; the estimate of it falls to zero. They cross near $\rho = 0.5$.

ρ	one member's error	ensemble's **true** error $\mathrm{KL}(P^* \,\|\, \bar{Q})$	ensemble's **estimate** (BALD)
0.00	0.1569	0.0148	0.1196
0.25	0.1569	0.0432	0.0907
0.50	0.1558	0.0750	0.0614
0.75	0.1540	0.1102	0.0313
0.90	0.1521	0.1330	0.0127
1.00	0.1491	0.1491	$< 10^{-18}$

Table 2: corr(score, true gain) in JUNCTION, by why the cheap forecaster is worse; negative is worse than routing at random. *n/a* marks a configuration where the estimate does not exist rather than being small: a single deterministic forecaster has no ensemble spread, so BALD is identically zero and induces no ranking. The irreducible floor $\mathbb{E}[H(P^*)]$ runs 0.72 → 1.33 nats across the columns.

	ranking by total uncertainty, $\tau =$					BALD	
why the cheap model is worse	0.35	0.70	1.30	2.00	3.50	τ=0.35	τ=3.5
missing evidence	+0.16	+0.13	+0.05	+0.08	+0.04	n/a	n/a
injected noise	+0.37	+0.24	−0.01	−0.25	−0.61	+0.33	−0.24
overconfidence (T=0.4)	−0.44	−0.75	−0.97	−1.00	−0.99	n/a	n/a
underconfidence (T=2.5)	−0.31	−0.72	−0.93	−0.97	−0.98	n/a	n/a
evidence ensemble	−0.28	−0.33	−0.45	−0.46	−0.46	+0.56	+0.66
mixed	+0.41	+0.29	+0.09	−0.10	−0.49	+0.35	−0.16

What that costs a router. At a fixed world, ranking scenes by the estimate correlates +0.114 with the true gain at $\rho = 0$ and −0.053 at $\rho = 0.75$; its budgeted skill against random goes +0.09 → −0.15. The advantage the estimator exists to provide is gone — and below random — before members are anywhere near identical. A single deterministic model is the $\rho = 1$ case in its most extreme and most common form: the estimate is identically zero and induces no ranking at all, including for a forecaster that ignores the prefix entirely and carries 0.386 nats of recoverable error.

The aleatoric half of the same estimate fails in the mirror-image way: the true irreducible term is 0.9897 nats for every forecaster, as (1) requires, while the estimate of it swings from 0.65 to 1.39 with the forecaster's confidence — two forecasters with identical accuracy and identical true irreducible uncertainty report 0.65 and 1.23 (Appendix B).

4 Whether confidence routing inverts depends on why the cheap model is worse

Sweeping the floor with one cheap/expensive pair invites the reading that the floor is the whole story. It is not. Table 2 fixes the pair's *cause* and sweeps the floor, with a random baseline at zero by construction.

Missing evidence is benign. A model that cannot see a cue is honestly uncertain, and that is also where a better model gains. The correlation decays toward zero (+0.158 → +0.043) but never inverts — literally toward zero, in that the high-floor end has a bootstrap interval of $[-0.001, +0.086]$, so the honest reading there is "no signal" rather than "a little signal".

Miscalibration is catastrophic, and it inverts at every floor. An overconfident model's entropy is lowest exactly where it is most wrong, so ranking by entropy systematically selects the scenes it already handles: −0.443 → −0.992. Underconfidence is nearly as bad (−0.307 → −0.978). This is the case that matters outside a synthetic world, because overconfidence is the documented failure mode of neural classifiers [4] rather than a hypothetical one. Xia et al. [8] document the same direction for label smoothing in selective classification; what the controlled floor adds is that the inversion is not a property of the regulariser but of the confidence–error relationship, and that it deepens monotonically as the irreducible floor rises.

Where the estimate is well-posed, it works, and it is floor-invariant. With members that disagree

Table 3: The two decisions are driven by almost disjoint quantities. Share is the fraction of the mean prefix-extension gain that is $\Delta H(P^*)$; the correlations are corr(total uncertainty, gain) for each of the two available actions.

	JUNCTION (additive evidence)			KITCHEN (non-additive)		
prefix L	$\Delta H(P^*)$ share	$\rightarrow$ model	$\rightarrow$ prefix	$\Delta H(P^*)$ share	$\rightarrow$ model	$\rightarrow$ prefix
$1 \rightarrow 2$	94%	-0.189	$+0.151$	97%	-0.490	$+0.059$
$2 \rightarrow 4$	94%	$+0.001$	$+0.135$	93%	-0.346	$+0.052$
$4 \rightarrow 8$	95%	$+0.084$	$+0.117$	96%	-0.220	$+0.083$
$8 \rightarrow 16$	92%	$+0.127$	$+0.070$	97%	-0.133	$+0.060$

because evidence is missing — errors independent by construction — BALD scores $+0.557 \rightarrow +0.663$ and does not degrade at all as the floor rises, while total uncertainty is worse than random everywhere in the same setting ($-0.277 \rightarrow -0.460$); in the non-additive world it is stronger still ($+0.819 \rightarrow +0.843$). This is the positive result, and it is what makes the negative ones worth reporting: the distinction in (1) is real and actionable, and the whole difficulty is in estimating it.

The mechanism is explicable rather than mysterious: high predictive entropy selects scenes where the ensemble has hedged toward uniform, hedging toward uniform is close to optimal precisely when the world is uncertain, and those are the scenes with the least recoverable error. Peale et al. [7] name an adjacent condition — a benefit "whenever reducible and irreducible uncertainty are not too correlated" — on a different axis: theirs is the correlation between the two terms, ours is the cause of the deficit, which a practitioner can diagnose directly by measuring calibration.

5 Video has a second lever, and it is made of the term that cancels

An uncertain forecast admits two responses: spend on the model, or spend on the evidence. The second is available in video and not in image classification. It does not obey (2), because a longer prefix changes P^* itself:

$$\text{gain}(\text{extend prefix } L \rightarrow L') = \big[H(P^*_L) + \mathrm{KL}(P^*_L \,\|\, Q)\big] - \big[H(P^*_{L'}) + \mathrm{KL}(P^*_{L'} \,\|\, Q)\big], \tag{3}$$

whose leading term is $H(P^*_L) - H(P^*_{L'})$ — exactly the quantity that cancels from model escalation. Masks are nested across lengths, since evidence does not disappear as one keeps watching, and the floor is enumerated rather than sampled at every length (1.2122 $\rightarrow$ 0.9932 nats as L goes $1 \rightarrow 32$).

The robust part is the share: 92%–97% of the prefix-extension gain is the term that cancels entirely from model escalation, at every length in both worlds. The two decisions are therefore driven by almost disjoint quantities, and a system that ranks both by one confidence score is answering the wrong question about one of them.

Whether the two correlations have *opposite signs* depends on the world, and the honest version is the narrower one. In KITCHEN they do at every length: total uncertainty correlates $-0.490 \rightarrow -0.133$ with the model decision and $+0.059 \rightarrow +0.060$ with the prefix decision throughout. In JUNCTION the opposition holds only at the shortest prefix (-0.189 against $+0.151$); by $4 \rightarrow 8$ the model correlation has turned positive. Short prefixes with interacting cues are the regime video forecasting operates in, and it is where the opposition is robust; the additive world is where it is not. Prior work on video anticipation under ambiguity treats the prediction *horizon* as the axis [1]; the observation prefix is a different lever, and the non-cancellation in (3) is what makes it a different decision.

6 Robustness

Every claim above is a claim about a sign, and a sign from one run is not yet a claim. We re-ran all three families over 5 seeds — re-drawing the scenes *and* the forecasters' noise streams — and separately bootstrapped the published run with 1000 resamples of the scenes, which keeps seed-to-seed variation apart from the precision of a single reported figure (Appendix E) — and for one cell the seed spread lies outside that run's bootstrap interval, which is the reason to report both rather than either. 24 of 24 (world, mechanism, floor) cells keep their published sign at every seed. The rise in the ensemble's true error as $\rho : 0 \rightarrow 1$ is 9.8–10.6$\times$ across seeds and worlds. BALD on the

well-posed ensemble averages $+0.563$ to $+0.840$ with a seed standard deviation of at most 0.012 and a bootstrap interval of $[+0.630, +0.694]$ at the high floor; the overconfidence inversion at the low floor sits in $[-0.485, -0.398]$. The prefix share is 92%–98% over every seed, world and length, and the prefix-decision correlation stays positive in 8 of 8 cells. One cell is not sign-stable and should not be read as a sign: the model-decision correlation keeps its sign in only 7 of 8, the exception being JUNCTION $2 \to 4$, whose published value $+0.001$ is indistinguishable from zero — the cell §5 already excludes.

7 What this is not evidence about

The world is synthetic, and that is not incidental. P^* exists here because we built it, so nothing above is evidence about any real corpus. That it transfers is an argument and not a measurement: the estimators under test are the deployed ones, so showing what they do where the truth is known is informative about what they do where it is not. A reader may decline the argument, and should then read every claim here as a claim about estimators only. **ρ is a knob, not a measurement:** we show what member correlation costs a router, not the ρ of any real ensemble, and whether repeated calls to one model form a high-ρ ensemble is an assumption — made more plausible by the removal of temperature controls from current model APIs, but not established here. **No vision-language model was called:** the arm is implemented, prompted, cached and costed at roughly one dollar, and no credential existed on the machine to run it. It is the only setting in which a real model could be handed a true $\mathrm{KL}(P^* \,\|\, Q)$, and its three predictions are recorded, falsifiable and untested.

8 Conclusion

The reducible/irreducible distinction is real and actionable: where the estimate is well-posed it ranks escalation candidates at $+0.557$ to $+0.843$ and does not care how uncertain the world is. The difficulty is entirely in estimating it, and the regime where the estimate inverts against the truth — members that share their errors — is the regime a deployment is in. Confidence routing's failure, meanwhile, is not a fact about confidence but about the cheap model's calibration, which is diagnosable in advance; and video's second lever is made almost entirely of the term that cancels from the first, so one score cannot serve both.

References

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A The instrument, and the two worlds

Every result in the paper is measured in two worlds that differ on every axis that could plausibly matter, so that a finding which depends on the construction can be told from one that does not. KITCHEN is the important one: its evidence is *non-additive*, so losing a cue can be catastrophic rather than gradual, which is the structure most likely to break a mechanism map built on gradual evidence loss. Cue visibilities are deliberately spread rather than uniform — if every cue were equally visible the observability mask would carry no information and every scene would sit at the same irreducible floor, which is the degenerate case a routing experiment must avoid.

Table 4: The two worlds. Only the second admits interactions between cues, and the instrument's checks are world-independent by construction rather than by configuration.

	JUNCTION	KITCHEN
outcomes	4	3
cues	4	5
cue alphabets	uniformly ternary	mixed binary and ternary
symmetry	left/right mirrored	none
evidence	**additive** (linear in the cues)	**non-additive** (pairwise)
joint cue states	81	108
irreducible floor at the default τ	0.9919 nats	0.7792 nats
validation checks passed	7/7	7/7

Table 5: The validation checks. Each expectation was fixed by the mathematics before the check ran, and every check is world-independent by construction — which is tested rather than asserted by running them in the second world, where they also pass.

	claim	expected	measured
V1	Bayes-optimal forecaster has zero epistemic uncertainty	max KL(P*‖Q) = 0 exactly	max = 0.000e+00
V2	Uniform forecaster scores log K on every scene	expected NLL = log 4 = 1.386294 exactly	max deviation = 4.441e-16
V3	Marginal forecaster's excess loss equals I(y; z_m)	I(y; z_m) = 0.383812 nats (exact)	0.382810 $\pm$ 0.014659 (3 sem)
V4	Expected epistemic error is monotone in the cues used, and matches closed form	ladder non-decreasing; every rung within 3 sem of its enumerated value	monotone=True; 5/5 rungs match
V5	Aleatoric floor is non-increasing in prefix length	every successive difference ≤ 0	max increase = -1.014e-02
V6	As members correlate, the estimate and the true error move in opposite directions	single-member error flat; KL(P*‖Qbar) rises monotonically; BALD falls monotonically to 0	single 0.1569→0.1491; true 0.0148→0.1491 (10.0x); BALD 0.1196→3.75e-19
V7	Oracle routing dominates every policy at every budget	no policy achieves lower loss than the oracle at any budget	largest violation = 0.000e+00

A renderer exists so that the vision-language arm would be a real experiment rather than a description of one; it is not used by any measurement in this paper. Its contract is that a cue the mask hides must not be recoverable from the pixels, which is verified on 682 scene pairs differing only in a hidden cue: maximum pixel difference exactly zero.

B The full decomposition table

Table 6: Both decompositions side by side, JUNCTION, τ=0.9. The **true** irreducible column is constant, as (1) requires. The **estimated** column (bold) is not: it tracks the forecaster's confidence. The estimated epistemic column is exactly zero for every deterministic forecaster, including one whose true epistemic error is the largest in the table among forecasters that read the prefix at all.

forecaster	irred. true	irred. **est.**	epist. true	epist. est.	acc.
Bayes-optimal	0.9897	**0.9897**	0.0000	0.0000	0.575
tempered T=0.5	0.9897	**0.6543**	0.1480	0.0000	0.575
tempered T=2	0.9897	**1.2348**	0.0746	0.0000	0.575
partial (2 of 4 cues)	0.9897	**1.1065**	0.1149	0.0000	0.522
partial (1 of 4 cues)	0.9897	**1.1769**	0.1848	0.0000	0.482
logit noise σ=0.75, ρ=0	0.9897	**0.9136**	0.0147	0.1206	0.562
marginal (ignores prefix)	0.9897	**1.3758**	0.3862	0.0000	0.290
uniform	0.9897	**1.3863**	0.3966	0.0000	0.290

C The mechanism map in the second world

Table 7: As Table 2, in KITCHEN: three outcomes, five cues, mixed alphabets, no symmetry, and *non-additive* evidence. No sign reverses between the worlds. Two findings sharpen: BALD on the well-posed ensemble improves, and the benign case stops decaying.

	ranking by total uncertainty, $\tau =$					BALD	
why the cheap model is worse	0.35	0.70	1.30	2.00	3.50	τ=0.35	τ=3.5
missing evidence	+0.19	+0.14	+0.18	+0.20	+0.17	n/a	n/a
injected noise	+0.31	+0.13	−0.12	−0.28	−0.57	+0.31	−0.24
overconfidence (T=0.4)	−0.22	−0.74	−0.97	−1.00	−1.00	n/a	n/a
underconfidence (T=2.5)	−0.72	−0.93	−0.98	−0.99	−1.00	n/a	n/a
evidence ensemble	−0.44	−0.54	−0.63	−0.64	−0.62	+0.82	+0.84
mixed	+0.31	+0.21	+0.03	−0.19	−0.50	+0.24	−0.20

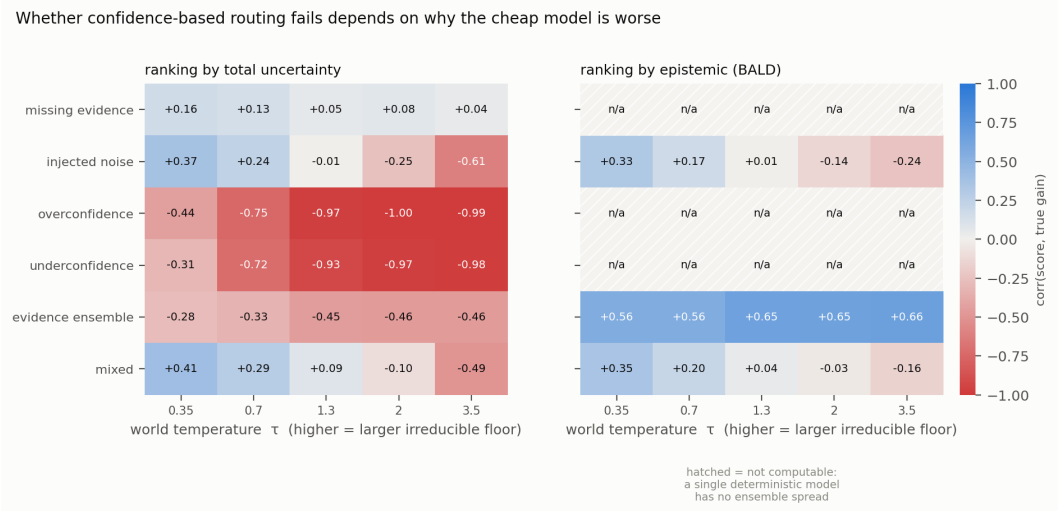


Figure 1: The mechanism map of Table 2 as a heat map. Hatched cells are configurations in which the estimate does not exist rather than being small.

D Budgeted curves and the skill score

The correlation reported throughout is a ranking diagnostic. The deployment-facing number is the budgeted escalation curve: at budget b the policy escalates the b fraction of scenes it scores highest, and the curve reports the resulting mean expected loss. Efficiency is normalised against the oracle *at the same budget*, since a policy at budget b cannot do better than escalating the b scenes with the largest true gain, and then rescaled so that random routing scores 0 and the oracle 1. Without that rescaling a random policy scores about 0.6 against the oracle, which compresses every comparison into the top of the range and makes a policy that is worse than random still look as though it captures most of the gain.

The floor sweep with a single cheap/expensive pair, for reference, since it is the setting in which the effect was first reported: ranking by total predictive uncertainty correlates $+0.336$ with the true gain at the lowest floor and -0.590 at the highest, with a budgeted skill of -0.61 there — that is, well below random. Ranking by the epistemic estimate falls less far over the same sweep ($+0.294$ to -0.223), and both pass through zero near $\tau = 1.3$. Peale et al. [7] report the same qualitative comparison against a pointwise-optimal policy on synthetic data; what §4 adds is that this setting is one row of a table whose other rows do not invert at all.

The correlation-sweep counterpart is in §3: the epistemic policy's budgeted skill against random goes $+0.09$ at $\rho = 0$ to -0.15 at $\rho = 0.75$, so the budgeted number and the rank correlation agree about where the estimator stops helping. Ties in a policy's scores are broken by index order rather than at random, which is the pessimistic choice for a policy that emits many identical scores — and that is the case worth being pessimistic about, since a policy which cannot separate scenes should not be rewarded for it.

E Robustness in full

Table 8: Seed spread over 5 seeds, which re-draw the scenes and the forecasters' noise streams, and the percentile bootstrap interval of the published run (1000 resamples of the scenes; the members are not a sample from a population, so they are not resampled). "sign" is the fraction of seeds agreeing with the published sign.

quantity	mean	sd	min, max	sign	bootstrap 95% CI
total unc. → gain, missing evidence, τ=3.5 (junction)	+0.025	0.019	+0.009, +0.046	100%	[−0.001, +0.086]
total unc. → gain, overconfidence, τ=3.5 (junction)	−0.993	0.001	−0.993, −0.991	100%	[−0.994, −0.990]
BALD → gain, evidence ensemble, τ=3.5 (junction)	+0.652	0.012	+0.638, +0.663	100%	[+0.630, +0.694]
aleatoric share of prefix gain, 1 → 2 (junction)	95%	0.007	94%, 95%	100%	—
total unc. → prefix gain, 1 → 2 (junction)	+0.145	0.008	+0.134, +0.156	100%	—
rise in KL($P^*\|\bar{Q}$), ρ : 0 → 1 (junction)	10.3×	0.206	10.0×, 10.6×	100%	—
total unc. → gain, missing evidence, τ=3.5 (kitchen)	+0.186	0.034	+0.135, +0.221	100%	[+0.127, +0.210]
total unc. → gain, overconfidence, τ=3.5 (kitchen)	−1.000	0.000	−1.000, −1.000	100%	[−1.000, −1.000]
BALD → gain, evidence ensemble, τ=3.5 (kitchen)	+0.840	0.007	+0.831, +0.849	100%	[+0.828, +0.858]
aleatoric share of prefix gain, 1 → 2 (kitchen)	97%	0.011	96%, 98%	100%	—
total unc. → prefix gain, 1 → 2 (kitchen)	+0.056	0.010	+0.046, +0.072	100%	—
rise in KL($P^*\|\bar{Q}$), ρ : 0 → 1 (kitchen)	9.9×	0.077	9.8×, 9.9×	100%	—

F Reproduction

Everything is CPU-only, single-process, and about ninety seconds end to end for the main results; the seed sweep in Appendix E takes a few minutes more. There is no GPU, no network access, no API key and no data file involved. Each experiment writes one row per scored prediction alongside the reduced numbers, so every figure in this paper traces to the rows it came from, and the paper's own numerals are generated from those files rather than typed.

```
python3 tests/test_instrument.py             # 23/23 identities
python3 scripts/validate_instrument.py       # 7/7 checks
python3 scripts/validate_instrument.py --world kitchen --tau 0.70
python3 scripts/run_decomposition.py         # Table 5
python3 scripts/run_routing.py               # Appendix D
python3 scripts/run_mechanisms.py            # Table 2
python3 scripts/run_mechanisms.py --world kitchen        # Table 6
python3 scripts/run_prefix.py                # Table 3
python3 scripts/run_prefix.py --world kitchen --tau 0.70
python3 scripts/run_robustness.py            # Table 7
python3 scripts/make_figures.py
python3 scripts/audit_numbers.py             # re-measures the docs
python3 scripts/make_paper_tables.py         # this paper's numbers
python3 scripts/audit_paper.py               # fails on any drift
```

Results land in `experiments/results/<name>/` as a `records.csv` (one row per scored prediction) and a `summary.json` (the reduced numbers), so every claim traces to the rows it came from. Seeds derive from one root through a hash of a label path, so adding an arm to an experiment cannot change the draws of the arms beside it.